Pranav **Kairon**

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Summary

I am a physics PhD student at UIUC working in quantum information and computational quantum many-body physics, with experience developing and benchmarking quantum algorithms for electronic structure and many-body problems. My research spans variational quantum algorithms, quantum Monte Carlo, tensor networks, and neural quantum states, with additional training in quantum error correction and fault-tolerant quantum computing. I'm experienced in Python-based scientific computing, Qiskit/JAX, Git, and high-performance computing, with a strong interest in quantum algorithm development, compilation, and resource-efficient implementations for fault-tolerant architectures.

Education

University of Illinois Urbana Champaign — *Urbana, USA*

PhD in Physics — *August 2024-Present*

- Working with Prof. Bryan Clark
- Teaching Assistant for PHYS 211 (Classical Mechanics)
- CGPA: 4.0/4

University of British Columbia — *Vancouver, Canada*

M.S. in Physics — *2021 – August 2024*

- Thesis: **From quantum computing to condensed matter : a machine learning perspective**
- Teaching Assistant for PHYS 100 (Classical Mechanics) and PHYS 131 (Electromagnetism).
- CGPA: 3.65/4.

Delhi Technological University — *New Delhi, India*

B.Tech. in Engineering Physics — *2017 – 2021*

- Thesis: **Coherence-based inequality for state discrimination**
- CGPA: 8.94/10.

Publications and Posters

Geometric View of Iterative Fixed-Node Dynamics — *Urbana, USA*

Conference Talk/Journal Article — *August 2026*

- Kairon, Pranav, and Bryan K. Clark. "Geometric View of Iterative Fixed-Node Dynamics." arXiv:2609.16308 (2026).

Extrapolation of polaron properties to low phonon frequencies by multi-fidelity Bayesian machine learning — *Vancouver, Canada*

Conference Talk/Journal Article — *August 2023*

- Presented a talk and poster at the RTGdyn Conference 2022.
- Kairon, P., Sous, J., Berciu, M., & Krems, R. V. (2024) Physical Review B, 109(14), 144523.

Equivalence between exponential concentration in quantum machine learning kernels and barren plateaus in variational algorithms — *Vancouver, Canada*

Journal Article — *August 2023*

- Kairon, Pranav, Jonas Jäger, and Roman V. Krems. Physical Review A 114.2 (2026): 022417.
- Presented a talk and poster at the RTGdyn Conference 2024.

Coherence-based inequality for the discrimination of three-qubit GHZ and W class — *Delhi, India*

Journal Article — *January 2022*

- Kairon, P., Singh, M., Adhikari, S. Coherence-based inequality for the discrimination of three-qubit GHZ and W class. Quantum Inf Process 21, 173 (2022)

Noisy three-player dilemma game: robustness of the quantum advantage — *Delhi, India*

Journal Article — *April 2020*

- Kairon, P., Thapliyal, K., Srikanth, R. et al. Noisy three-player dilemma game: robustness of the quantum advantage. Quantum Inf Process 19, 327 (2020).

COVID-19 Outbreak Prediction Using Quantum Neural Networks — *DoSIER 2020 (Virtual)*

Conference Paper — *June 2021*

- Kairon, P., Bhattacharyya, S. (2021). COVID-19 Outbreak Prediction Using Quantum Neural Networks. In: Bhattacharyya, S., Dutta, P., Datta, K. (eds) Intelligence Enabled Research. Advances in Intelligent Systems and Computing, vol 1279. Springer, Singapore.

Work Experience

Iliad
USA

Iliad Intensive
October 2026 – November 2026

- Selected for four-week training program focused on foundational, mathematically grounded AI alignment research

Mitacs Accelerate
Canada

Research Intern
October 2023 – August 2024

- Using quantum machine learning methods to perform feature selection for a metabolomics dataset with millions of dimensions.
- Collaborated on a grant proposal with a supervisor and a company, securing $30k CAD in funding.

Superconducting Circuit Design and Fabrication by CMC Microsystems
Canada

Workshop Participant
June 2022 – July 2022

- Designed, simulated, and fabricated superconducting circuits, including resonators, SQUIDs, and qubits.
- Conducted advanced testing of superconducting chips using a 4K probe station and a dilution refrigerator.
- Gained hands-on experience with low-temperature experimental techniques for quantum information science.

HCL Technologies
Bengaluru, India

Research Intern
August 2021 – September 2021

- Benchmarked methods for solving the electronic structure problem on DWave quantum annealer (AWS Amazon Braket).
- Explored error mitigation schemes for variational quantum eigensolvers using Hamiltonian symmetries.

Conduit Computing
Boston, USA (Remote)

Research Intern
March 2020 – January 2021

- Simulated proteins for COVID-19 drug development using NAMD and VMD on the Frontera supercomputer.
- Proposed research selected for the White House HPC consortium and Nvidia Supercomputers.
- Presented a full research proposal at Quantum British Columbia in a virtual poster session.

Qnu Labs (India's First Quantum Company)
Bengaluru, India

Research Intern
December 2019 – January 2020

- Developed MATLAB/Mathematica programs to calculate key rate deterioration with distance for DPS QKD protocols.
- Simulated high-throughput quantum random number generators based on LASER phase noise.

Solid State Physics Laboratory – DRDO
Delhi, India

Research Intern
May 2019 – August 2019

- Characterized materials using Scanning Tunnelling Microscopy and fabricated MIS structures for AlN thin films.
- Repaired a table top STM, reconfiguring software and hardware setups independently.

Solid State Physics Laboratory – DRDO
Delhi, India

Research Intern
December 2018 – February 2019

- Modeled thin film characterization using X-ray reflectivity with MATLAB.
- Determined composition, stress, and strain of AlGaN/GaN multilayer structures for HEMT devices.

Laser Science and Technology Center – DRDO
Delhi, India

Research Intern
May 2018 – August 2018

- Developed an optical resonator for high-power lasers using Nd-doped $PoCl_3$ as the gain medium.
- Surveyed various optical coatings for laser-grade mirrors.

Skills

Programming Languages	Python, C++, MATLAB, Mathematica,R
Packages and Frameworks	Qiskit, Pennylane, IBMQ Experience, JAX, TensorFlow, PyTorch, ITensor, dmrgpp
Utilities and Tools	Git/GitHub, Linux Shell, VSCode, Jupyter Notebook, SQL, MongoDB

Extracurricular Activities

NSERC Quantum Computing CREATE
Vancouver, Canada

Mentor and Judge
November 2022

- Mentored two groups of high school students on using quantum computing for interplanetary communication.
- Judged final presentations for the high school engineering and design competition.

Deltech Engineering Physics Technological Hub (DEPTH)
Delhi Technological University, India

Vice-President and Lecturer
2017 – 2021

- Organized seminars, debates, guest lectures, and paper presentations to enhance students' understanding of various scientific fields.
- Conducted lectures on advanced topics not covered in university courses and mentored beginners in systematic exploration of these topics.

Student Mentorship Programme — *Delhi Technological University, India*

MENTOR — *2019 – 2021*

- Mentored five freshmen from the Engineering Physics Department.
- Provided guidance on both personal and academic challenges throughout the year.

Honors, Grants, & Scholarships

2023	**Mitacs Accelerate Grant**, Secured funding for a 10-month internship in collaboration with a biomedical startup to explore quantum machine learning applications.	*Canada*
2022	**IBM Error Correction Summer School**, Selected for in-person IBM summer school in New York from hundreds of applicants.	*New York, USA*
2022	**Commendable Research Award at DTU**, Received the award and a cash prize of Rs. 50k for publishing a journal article during my bachelor's thesis.	*New Delhi, India*
2022	**Quantum BC Roadmapping Workshop Winner**, Won first place in a workshop developing a quantum-based solution for an industrially relevant genomics problem.	*Canada*
2021	**QuEST Fellowship**, Awarded by the Stewart Blusson Quantum Matter Institute, University of British Columbia.	*Vancouver, Canada*
2021	**NSERC Quantum Computing CREATE Scholarship**, Received prestigious funding for research at UBC.	*Vancouver, Canada*
2021	**MCQST Summer School Selection**, Shortlisted among 20 students out of 500 applicants for the Munich Center of Quantum Science and Technology Summer School.	*Munich, Germany (Virtual)*
2020	**Cornerstone Models of Quantum Computing Summer School**, Selected for the summer school conducted by Quantum British Columbia, among 130 undergraduate and graduate candidates.	*Canada (Virtual)*
2017	**All India Rank - 4922, JEE Main**, Achieved this rank in the JEE Main examination, taken by approximately 1.5 million students.	*India*

References

1	**Prof. Bryan K. Clark**, University of Illinois Urbana Champaign, USA	*Email*
2	**Prof. Roman Krems**, University of British Columbia, Canada	*Email*
3	**Prof. Anirban Pathak**, Jaypee Institute of Information Technology, India	*Email*